

# Practice Midterm Solutions

GSE 510

Total Points: 100  
10 Questions, 10 Points Each

**Instructions:** These are worked solutions for the practice midterm.

## 1. Sets and Logic

(10 points)

Let

$$A = \{x \in \mathbb{R} : 0 \leq x \leq 10\}, \quad B = \{x \in \mathbb{R} : x > 5\}, \quad C = \{x \in \mathbb{R} : x < 8\}.$$

- (a) Since  $A$  contains all real numbers between 0 and 10, including the endpoints, and  $B$  contains all real numbers greater than 5,

$$A \cap B = \{x \in \mathbb{R} : 5 < x \leq 10\} = (5, 10].$$

- (b) Since  $C$  contains all real numbers less than 8,

$$A \cap C = \{x \in \mathbb{R} : 0 \leq x < 8\} = [0, 8).$$

- (c) We need  $x > 5$  and  $x < 8$ , so

$$B \cap C = \{x \in \mathbb{R} : 5 < x < 8\} = (5, 8).$$

- (d) Combining all three restrictions gives

$$A \cap B \cap C = \{x \in \mathbb{R} : 5 < x < 8\} = (5, 8).$$

- (e) If the universal set is  $\mathbb{R}$ , then the complement of  $A$  is the set of all real numbers not in  $A$ :

$$A^c = (-\infty, 0) \cup (10, \infty).$$

## 2. Bounds, Suprema, and Infima

(10 points)

Consider the set

$$S = \{x \in \mathbb{R} : 2 < x < 7\}.$$

- (a) Yes,  $S$  is bounded above. For example, 7 is an upper bound, and so is any number greater than 7.

- (b) Yes,  $S$  is bounded below. For example, 2 is a lower bound, and so is any number less than 2.

- (c) The least upper bound is

$$\sup S = 7.$$

- (d) The greatest lower bound is

$$\inf S = 2.$$

- (e) Neither 2 nor 7 belongs to  $S$ , because  $S = (2, 7)$  is an open interval. Therefore,

$$2 \notin S \quad \text{and} \quad 7 \notin S.$$

## 3. Sequences and Limits

(10 points)

For each sequence, determine whether it converges. If it converges, find its limit.

(a)

$$x_n = 5 + \frac{3}{n}.$$

Since  $\frac{3}{n} \rightarrow 0$  as  $n \rightarrow \infty$ ,

$$\lim_{n \rightarrow \infty} x_n = 5.$$

Thus, the sequence converges to 5.

(b)

$$y_n = \frac{2n+1}{n} = 2 + \frac{1}{n}.$$

Since  $\frac{1}{n} \rightarrow 0$ ,

$$\lim_{n \rightarrow \infty} y_n = 2.$$

Thus, the sequence converges to 2.

(c)

$$z_n = (-1)^n.$$

This sequence alternates between  $-1$  and  $1$ . It does not get close to a single limiting value. Therefore, it does not converge.

(d)

$$w_n = \frac{1}{n^2}.$$

Since  $n^2 \rightarrow \infty$ ,

$$\lim_{n \rightarrow \infty} w_n = 0.$$

Thus, the sequence converges to 0.

#### 4. Open and Closed Sets

(10 points)

Consider

$$A = (0, 1), \quad B = [0, 1], \quad C = (0, 1].$$

- (a)  $A = (0, 1)$  is open in  $\mathbb{R}$  because every point in  $A$  has a small open interval around it contained in  $A$ . It is not closed because it does not contain its boundary points 0 and 1.
- (b)  $B = [0, 1]$  is closed because it contains its boundary points 0 and 1. It is not open in  $\mathbb{R}$  because no open interval around 0 is fully contained in  $B$ , and similarly no open interval around 1 is fully contained in  $B$ .
- (c)  $C = (0, 1]$  is neither open nor closed in  $\mathbb{R}$ . It is not open because it contains 1, but no open interval around 1 is fully contained in  $C$ . It is not closed because it does not contain the boundary point 0.
- (d) The boundary points of  $A = (0, 1)$  are

$$0 \quad \text{and} \quad 1.$$

- (e) One example is

$$x_n = \frac{1}{n}.$$

For every  $n \geq 2$ , we have  $x_n \in (0, 1)$ , but

$$\lim_{n \rightarrow \infty} x_n = 0,$$

and  $0 \notin (0, 1)$ .

#### 5. Linear Algebra and Systems of Equations

(10 points)

Consider the system

$$\begin{aligned} 2x_1 + x_2 &= 11, \\ x_1 + 3x_2 &= 18. \end{aligned}$$

(a) In matrix form, this is

$$\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 11 \\ 18 \end{bmatrix}.$$

Thus,

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 11 \\ 18 \end{bmatrix}.$$

(b) The determinant is

$$|\mathbf{A}| = \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} = 2(3) - 1(1) = 6 - 1 = 5.$$

(c) Since

$$|\mathbf{A}| = 5 \neq 0,$$

the matrix  $\mathbf{A}$  is nonsingular.

(d) Use the inverse of  $\mathbf{A}$  and solve  $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$

(e) Nonsingularity implies that the system has a unique solution.

## 6. Span and Linear Independence in $\mathbb{R}^2$

(10 points)

Let

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

(a) A general linear combination is

$$a\mathbf{v}_1 + b\mathbf{v}_2 = a \begin{bmatrix} 2 \\ 1 \end{bmatrix} + b \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

Therefore,

$$a\mathbf{v}_1 + b\mathbf{v}_2 = \begin{bmatrix} 2a - b \\ a + 2b \end{bmatrix}.$$

(b) We want to know whether there exist  $a, b \in \mathbb{R}$  such that

$$\begin{bmatrix} 2a - b \\ a + 2b \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

This gives the system

$$2a - b = 3, \quad a + 2b = 4.$$

From the first equation,

$$b = 2a - 3.$$

Substitute into the second equation:

$$a + 2(2a - 3) = 4.$$

Hence,

$$a + 4a - 6 = 4,$$

so

$$5a = 10.$$

Therefore,

$$a = 2.$$

Then

$$b = 2(2) - 3 = 1.$$

Thus,

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix} = 2\mathbf{v}_1 + \mathbf{v}_2.$$

(c) The matrix whose columns are  $\mathbf{v}_1$  and  $\mathbf{v}_2$  is

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}.$$

Its determinant is

$$|A| = 2(2) - (-1)(1) = 4 + 1 = 5.$$

(d) Since

$$|A| = 5 \neq 0,$$

the vectors  $\mathbf{v}_1$  and  $\mathbf{v}_2$  are linearly independent.

(e) Since two linearly independent vectors in  $\mathbb{R}^2$  span  $\mathbb{R}^2$ , the vectors  $\mathbf{v}_1$  and  $\mathbf{v}_2$  span  $\mathbb{R}^2$ .

## 7. Linear Regression in Matrix Form

(10 points)

Suppose we estimate

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

using

$$\mathbf{y} = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}.$$

(a) The regression model in matrix form is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u},$$

where

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}.$$

(b) We have

$$\mathbf{X}' = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}.$$

Therefore,

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}.$$

(c) Also,

$$\mathbf{X}'\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 11 \\ 25 \end{bmatrix}.$$

(d) The OLS estimator is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

First,

$$\begin{vmatrix} 3 & 6 \\ 6 & 14 \end{vmatrix} = 3(14) - 6(6) = 42 - 36 = 6.$$

Therefore,

$$(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}.$$

Hence,

$$\hat{\boldsymbol{\beta}} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix} \begin{bmatrix} 11 \\ 25 \end{bmatrix}.$$

Multiplying,

$$\begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix} \begin{bmatrix} 11 \\ 25 \end{bmatrix} = \begin{bmatrix} 154 - 150 \\ -66 + 75 \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \end{bmatrix}.$$

Thus,

$$\hat{\boldsymbol{\beta}} = \frac{1}{6} \begin{bmatrix} 4 \\ 9 \end{bmatrix} = \begin{bmatrix} 2/3 \\ 3/2 \end{bmatrix}.$$

(e) The fitted regression line is

$$\hat{y} = \frac{2}{3} + \frac{3}{2}x.$$

## 8. Differentiation and Economic Interpretation

(10 points)

Suppose

$$C(q) = 50 + 10q + q^2.$$

(a) The marginal cost function is

$$C'(q) = 10 + 2q.$$

(b) At  $q = 20$ ,

$$C'(20) = 10 + 2(20) = 50.$$

(c) The second derivative is

$$C''(q) = 2.$$

(d) Since

$$C''(q) = 2 > 0,$$

marginal cost is increasing in  $q$ .

(e) A linear approximation gives

$$\Delta C \approx C'(20)\Delta q.$$

Since output rises from 20 to 21,

$$\Delta q = 1.$$

Therefore,

$$\Delta C \approx 50(1) = 50.$$

So the cost increase is approximately 50.

## 9. Approximation

(10 points)

Suppose

$$Y = F(K) = 10K^{1/2}.$$

The current level of capital is  $K_0 = 100$ .

(a) The first derivative is

$$F'(K) = 10 \cdot \frac{1}{2}K^{-1/2} = 5K^{-1/2}.$$

(b) The second derivative is

$$F''(K) = 5 \left( -\frac{1}{2} \right) K^{-3/2} = -\frac{5}{2}K^{-3/2}.$$

(c) First compute

$$F(100) = 10(100)^{1/2} = 10(10) = 100,$$

and

$$F'(100) = 5(100)^{-1/2} = \frac{5}{10} = \frac{1}{2}.$$

Since

$$104 - 100 = 4,$$

the linear approximation gives

$$F(104) \approx F(100) + F'(100)(104 - 100).$$

Thus,

$$F(104) \approx 100 + \frac{1}{2}(4) = 102.$$

(d) We have

$$F''(100) = -\frac{5}{2}(100)^{-3/2}.$$

Since

$$100^{3/2} = 1000,$$

we get

$$F''(100) = -\frac{5}{2} \cdot \frac{1}{1000} = -\frac{1}{400}.$$

The quadratic approximation is

$$F(K) \approx F(100) + F'(100)(K - 100) + \frac{1}{2}F''(100)(K - 100)^2.$$

Therefore,

$$F(104) \approx 100 + \frac{1}{2}(4) + \frac{1}{2}\left(-\frac{1}{400}\right)(4)^2.$$

Hence,

$$F(104) \approx 100 + 2 - \frac{8}{400} = 102 - 0.02 = 101.98.$$

(e) The quadratic approximation should be closer because it includes curvature through the second derivative. The exact value is

$$F(104) = 10\sqrt{104} \approx 101.9804,$$

so the quadratic approximation 101.98 is very close.

## 10. Implicit Differentiation and Comparative Statics

(10 points)

Suppose market equilibrium is defined by

$$D(p, a) = S(p),$$

where

$$D(p, a) = a - 2p$$

and

$$S(p) = 10 + p.$$

(a) The equilibrium condition is

$$a - 2p = 10 + p.$$

Bringing all terms to one side, define

$$F(p, a) = a - 2p - 10 - p.$$

Simplifying,

$$F(p, a) = a - 10 - 3p.$$

Therefore, equilibrium is characterized by

$$F(p, a) = 0.$$

(b) Solve explicitly:

$$a - 10 - 3p = 0.$$

Then

$$3p = a - 10.$$

Hence,

$$p^*(a) = \frac{a - 10}{3}.$$

(c) Differentiating the explicit solution,

$$\frac{dp^*}{da} = \frac{1}{3}.$$

(d) Using the implicit function theorem,

$$\frac{dp^*}{da} = -\frac{F_a}{F_p}.$$

Since

$$F(p, a) = a - 10 - 3p,$$

we have

$$F_a = 1 \quad \text{and} \quad F_p = -3.$$

Therefore,

$$\frac{dp^*}{da} = -\frac{1}{-3} = \frac{1}{3}.$$

This is the same answer as in part (c).

(e) The derivative

$$\frac{dp^*}{da} = \frac{1}{3}$$

means that if the demand-shift parameter  $a$  increases by one unit, the equilibrium price increases by approximately  $\frac{1}{3}$  units. Economically, an outward shift in demand raises the equilibrium price.